

ROLL NO.

291

**G.L. BAJAJ INSTITUTE OF TECHNOLOGY & MANAGEMENT
GREATER NOIDA**

B.Tech (CSE, CSDS, CS-AIML, CSIT, CSAI) 2nd Year

Sessional Test- 1 (ODD SEM 2026-27)

Subject Name: COMPUTER ORGANIZATION AND ARCHITECTURE

Subject Code: BCS302

Duration: 2:00 Hrs

Max. Marks: 50

- Note:
- (i) No student will be allowed to leave the examination Room before end of the exam.
 - (ii) Diagrams should be neat and clean.
 - (iii) Mention question number/section correctly.
 - (iv) Be precise in your answer.
 - (v) Do not write anything on question paper except Roll number.

Course Outcomes:

Following are the course outcomes of the subject :-

CO Code	Course Outcome (CO)	Bloom's Level
BCS302.1	Study of the basic structure and operation of a digital computer system.	K1(Remember), K2(Understand)
BCS302.2	Analysis of the design of arithmetic & logic unit and understanding of the fixed point and floating-point arithmetic operations.	K2(Understand), Applying(K4)
BCS302.3	Implementation of control unit techniques and the concept of Pipelining	Comprehending(K3),
BCS302.4	Understanding the hierarchical memory system, cache memories and virtual memory	K2(Understand)
BCS302.5	Understanding the different ways of communicating with I/O devices and standard I/O interfaces	K2(Understand), Applying(K4)

Section: A

(5*2 = 10)				
1. Attempt all questions.				
Q. No.	Questions	Marks	CO	BL
a)	Draw the basic functional units of computer.	2	BCS302.1	K2
b)	What is a bus? Name and briefly explain its types.	2	BCS302.1	K2
c)	Differentiate between general-purpose registers and special-purpose registers.	2	BCS302.1	K2
d)	Explain the need for a Look-Ahead Carry Adder.	2	BCS302.2	K2
e)	Define instruction cycle.	2	BCS302.1	K2

Section: B

(4*5 = 20)				
2. Attempt any four of the following:				
Q. No.	Questions	Marks	CO	BL
a)	Discuss the process of data transfer between registers, buses, and memory units in a digital system. Include explanations of read and write operation?	5	BCS302.1	K2
b)	Explain Daisy Chaining with advantages and disadvantages. Explain the role of DMA in Bus Arbitration.	5	BCS302.2	0
c)	Draw DFD of Restoring Division Method.	5	BCS302.2	K2
d)	Explain the IEEE-754 representation of floating-point numbers. Describe the significance of the sign, exponent, bias, and fraction fields in single-precision representation.	5	BCS302.2	K3
e)	Write short note on look ahead carry adders.	5	BCS302.2	K3
f)	Explain the execution of a complete Instruction Cycle.	5	BCS302.3	K3

Section: C

(1*10 = 10)

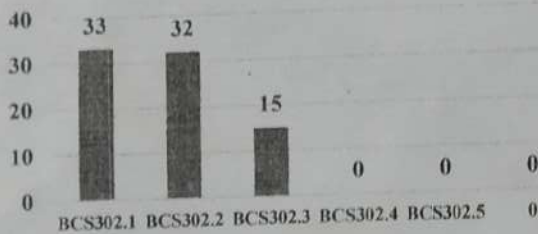
Attempt any one question:

Q. No.	Questions	Marks	CO	BL
a)	Explain various addressing modes with suitable examples. Given PC = 200, R1 = 400, XR = 100 and address field = 500, with specified memory contents, for each possible mode calculate the effective address and the operand that must be loaded into AC.	10	BCS302.1	K3
b)	Multiply the two numbers $(-10) \times (+2)$ by using the Booth's multiplication algorithm. Show the step by step multiplication.	10	BCS302.2	K3

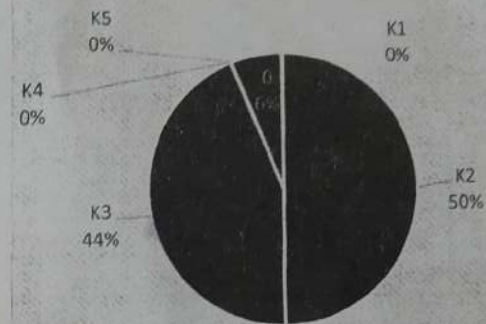
Section: D

4. Attempt any one question:		(1*10 = 10)		
Q. No.	Questions	Marks	CO	BL
a)	Explain different addressing modes with suitable examples and compare their effective-address calculation.	10	BCS302.1	K2
b)	Explain RISC & CISC architecture and its characteristics.	10	BCS302.3	K2

Course Outcome Wise Marks Distribution



Bloom's Level Distribution



Checked By
(Head of the Department)